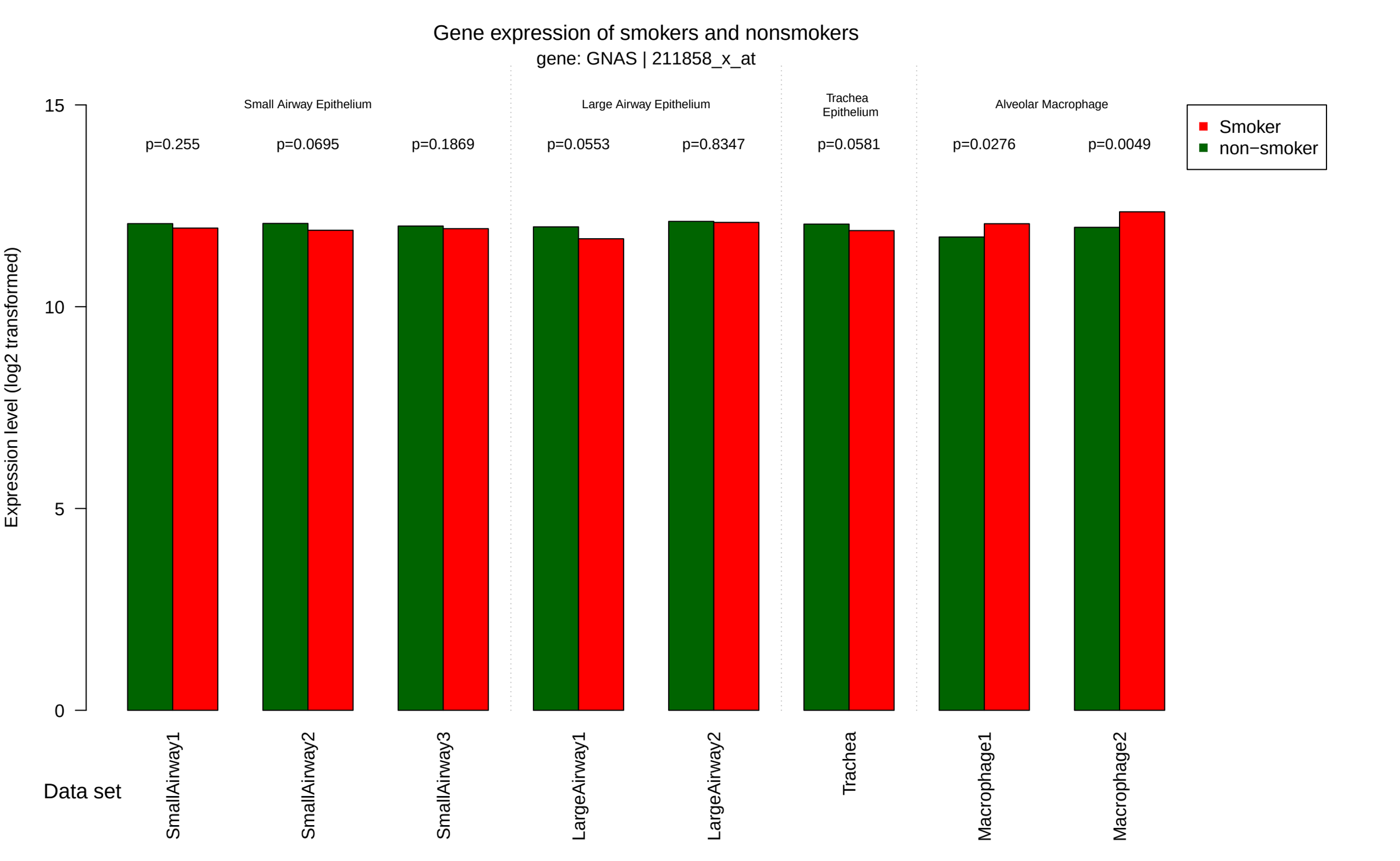
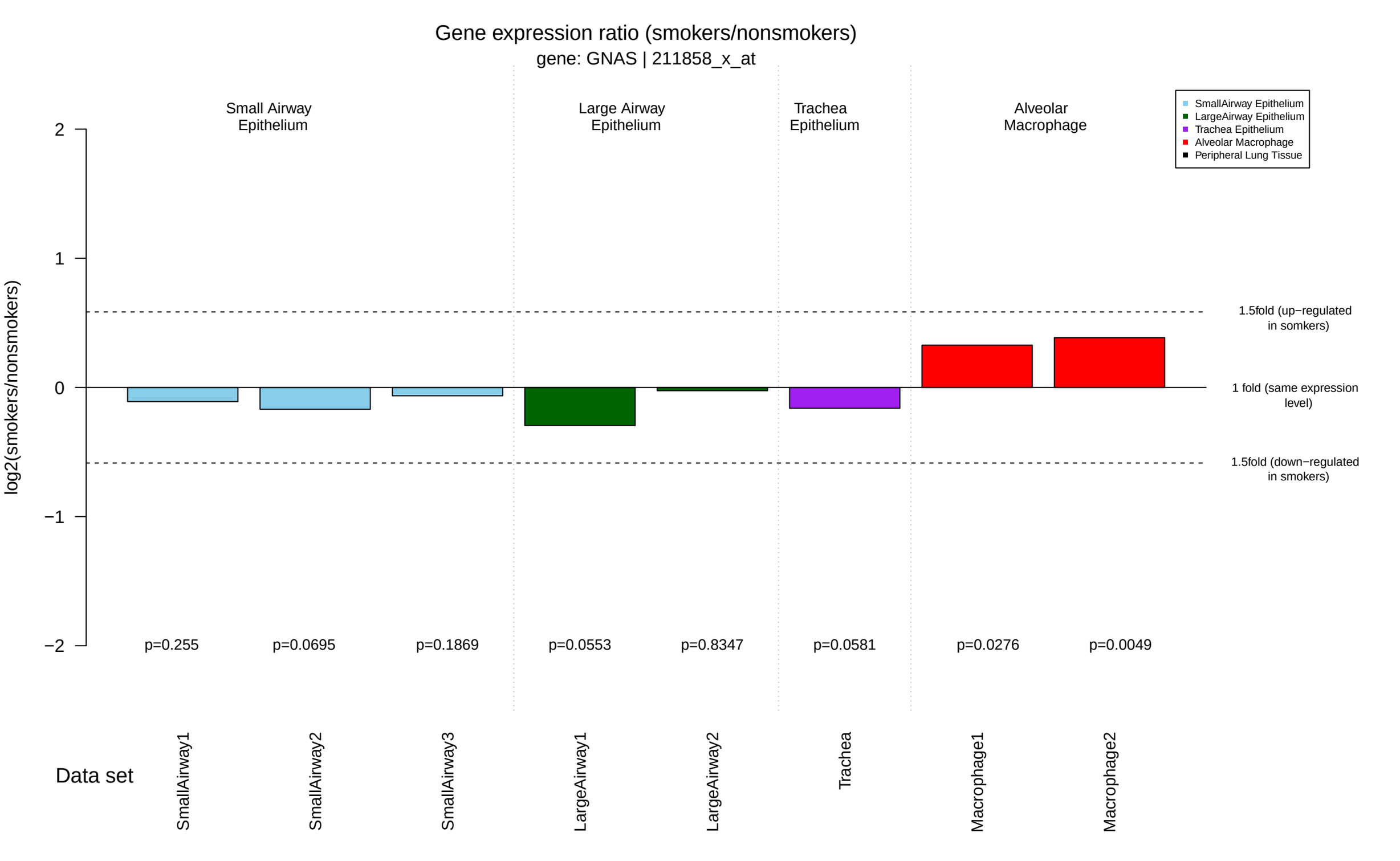
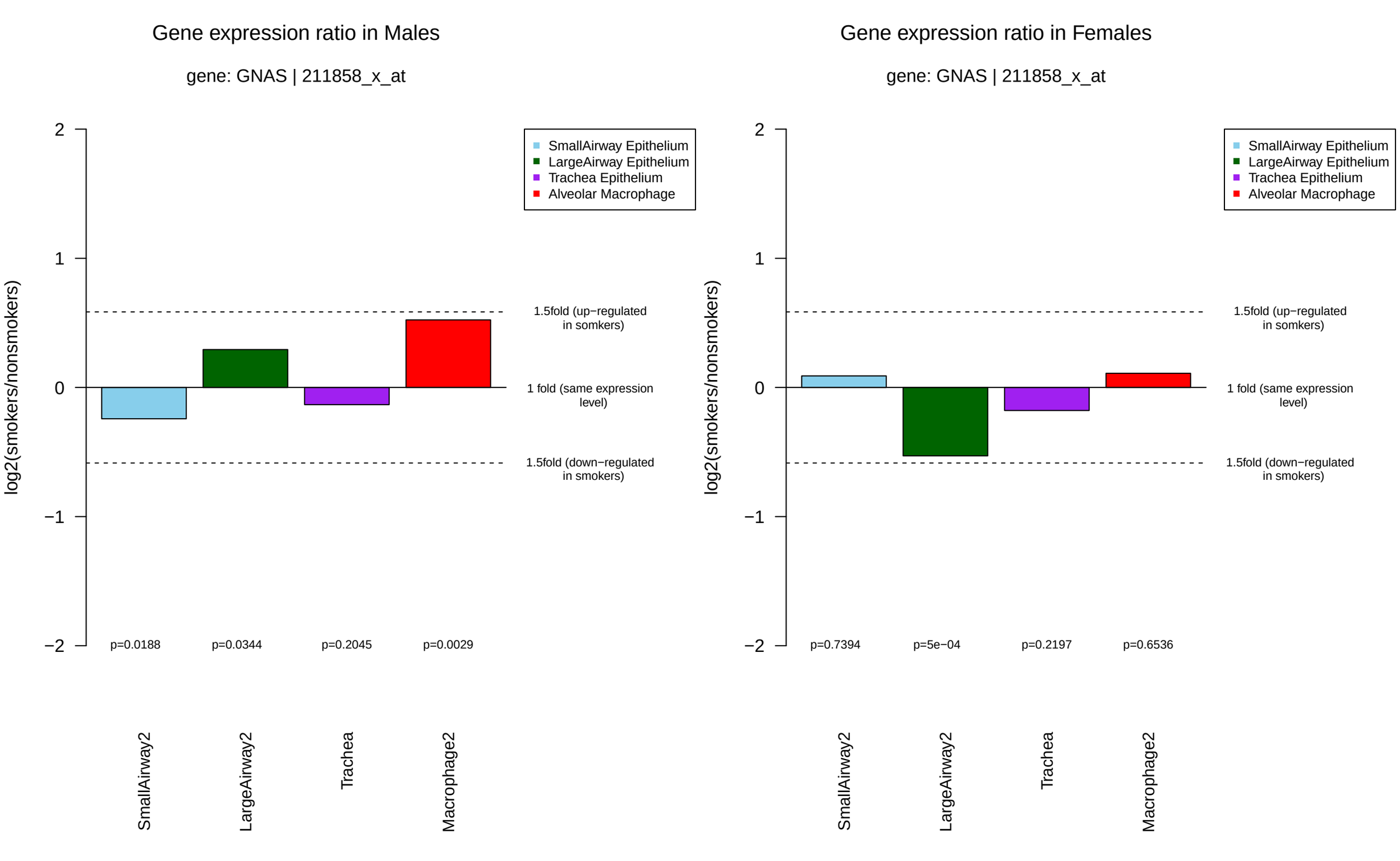
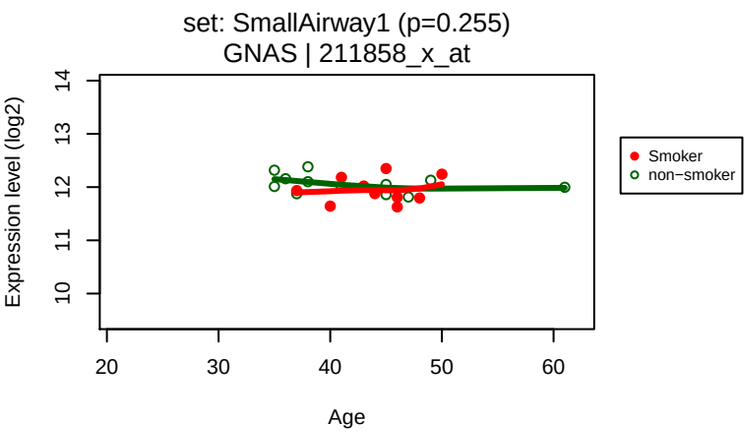
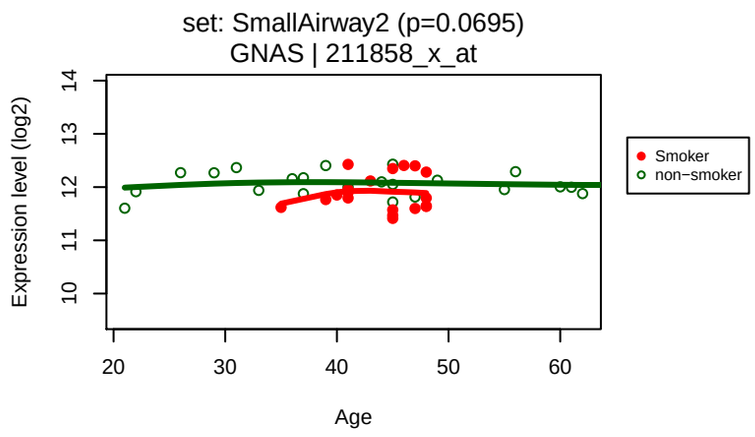
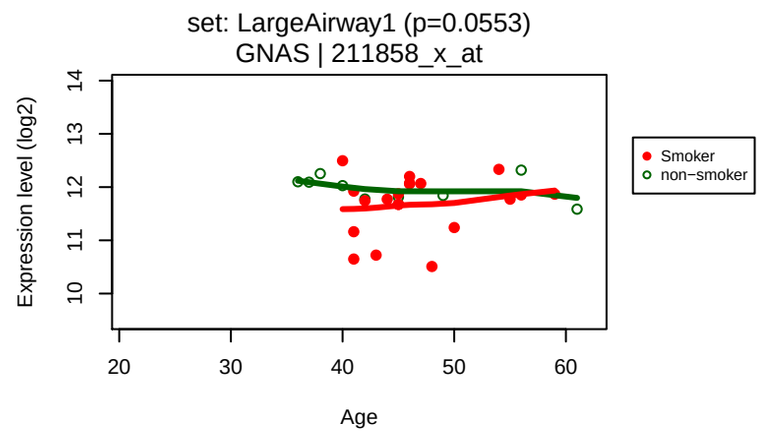

Gene expression of smokers and nonsmokers
gene: GNAS | 211858_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.255
p=0.0695
p=0.1869
p=0.0553
p=0.8347
p=0.0581
p=0.0276
p=0.0049
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

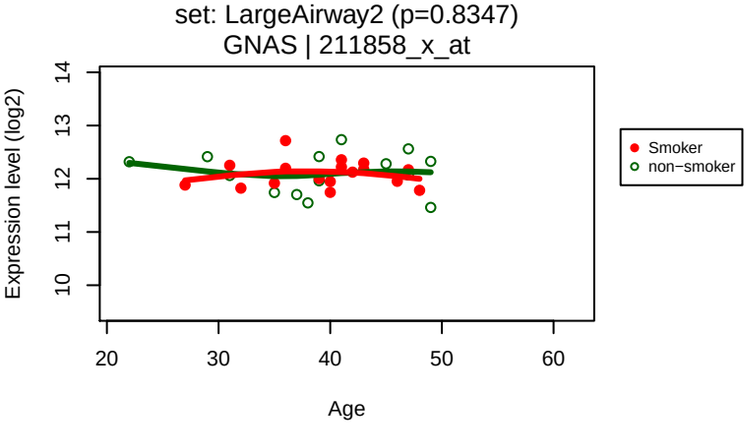

Gene expression ratio (smokers/nonsmokers)
gene: GNAS | 211858_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.255
p=0.0695
p=0.1869
p=0.0553
p=0.8347
p=0.0581
p=0.0276
p=0.0049
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

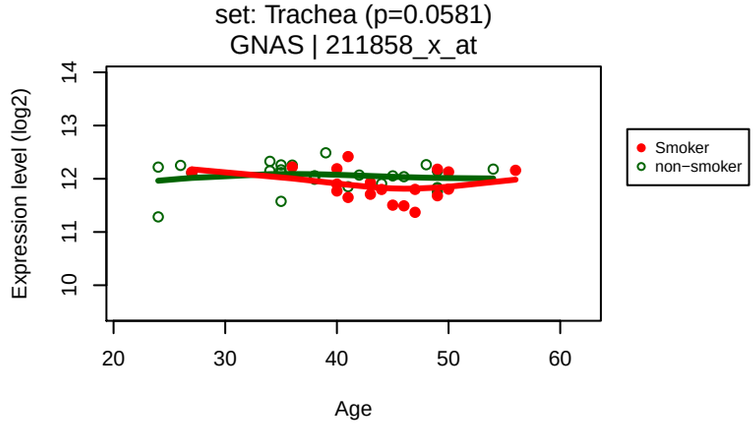

Gene expression ratio in Males
gene: GNAS | 211858_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0188
p=0.0344
p=0.2045
p=0.0029
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GNAS | 211858_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7394
p=5e−04
p=0.2197
p=0.6536
SmallAirway2
LargeAirway2
Trachea
Macrophage2

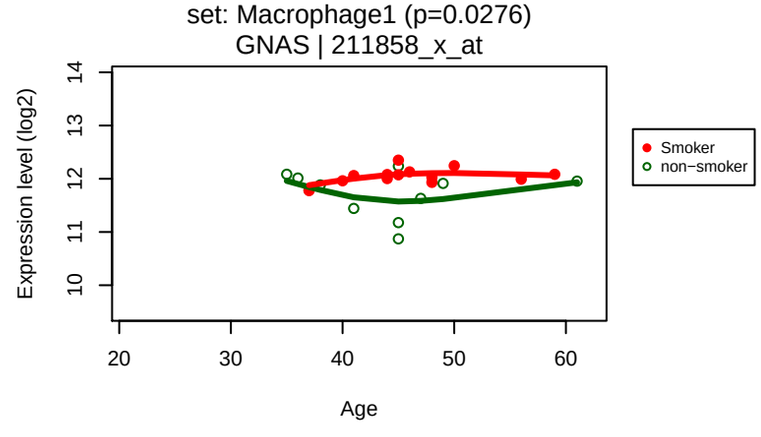

set: SmallAirway1 (p=0.255)
GNAS | 211858_x_at
Expression level (log2)
Age
Smoker
non−smoker

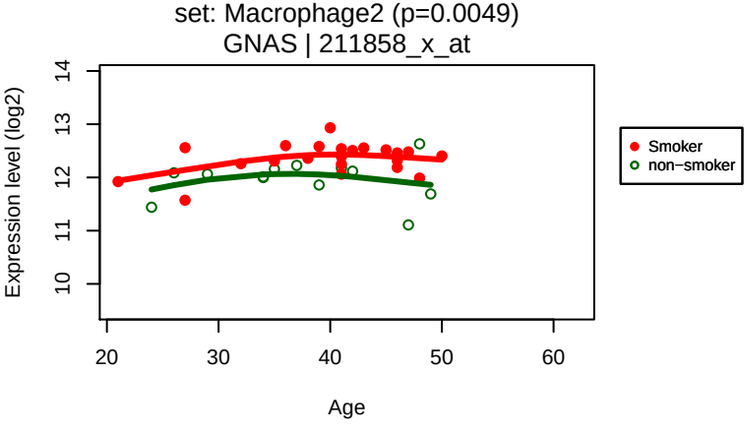

set: SmallAirway2 (p=0.0695)
GNAS | 211858_x_at
Expression level (log2)
Age
Smoker
non−smoker

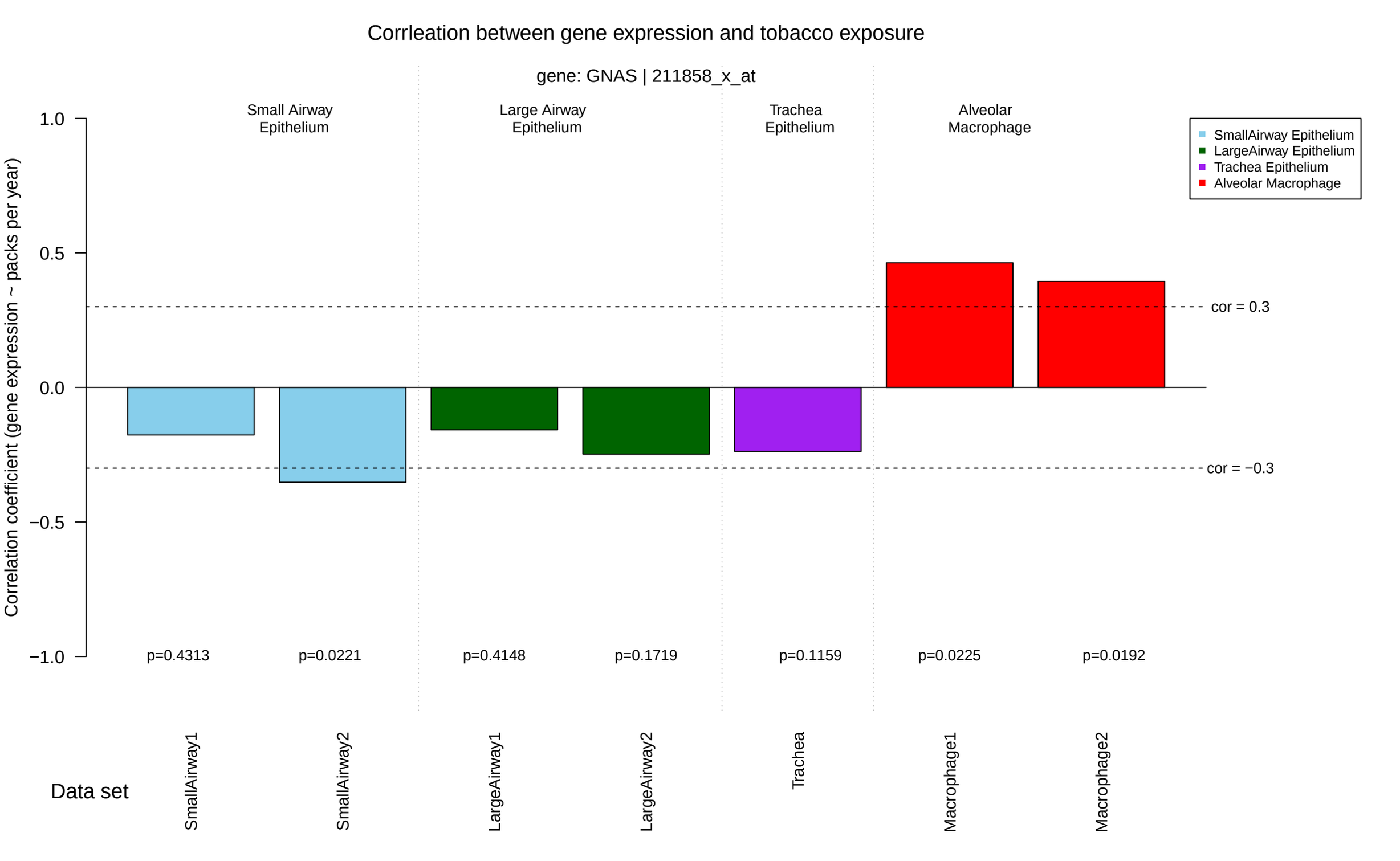

set: LargeAirway1 (p=0.0553)
GNAS | 211858_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8347)
GNAS | 211858_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0581)
GNAS | 211858_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0276)
GNAS | 211858_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0049)
GNAS | 211858_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GNAS | 211858_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4313
p=0.0221
p=0.4148
p=0.1719
p=0.1159
p=0.0225
p=0.0192
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2